

Proof of an Integral Inequality

Problem Statement

Prove that:

$$\frac{\pi^2}{9} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{2\pi^2}{9}.$$

Proof

Key Lemma: Monotonicity of $x/\sin x$ on $(0, \pi)$

Lemma 1. *The function $g(x) = \frac{x}{\sin x}$ is strictly increasing on $(0, \pi)$.*

Proof. Differentiating,

$$g'(x) = \frac{\sin x - x \cos x}{\sin^2 x}.$$

It suffices to show the numerator is positive. Define $\phi(x) = \sin x - x \cos x$. Then $\phi(0) = 0$ and

$$\phi'(x) = \cos x - \cos x + x \sin x = x \sin x > 0 \quad \text{for all } x \in (0, \pi),$$

since both $x > 0$ and $\sin x > 0$ on that interval. Hence $\phi(x) > 0$ on $(0, \pi)$, so $g'(x) > 0$, and g is strictly increasing on $(0, \pi)$. $\square$

Step 1: Bounds on the integrand

Since $[\pi/6, \pi/2] \subset (0, \pi)$ and g is strictly increasing on $(0, \pi)$ by Lemma 1, its minimum and maximum on $[\pi/6, \pi/2]$ are attained at the left and right endpoints respectively:

$$g\left(\frac{\pi}{6}\right) = \frac{\pi/6}{\sin(\pi/6)} = \frac{\pi/6}{1/2} = \frac{\pi}{3},$$

$$g\left(\frac{\pi}{2}\right) = \frac{\pi/2}{\sin(\pi/2)} = \frac{\pi/2}{1} = \frac{\pi}{2}.$$

Therefore, for all $x \in [\pi/6, \pi/2]$:

$$\frac{\pi}{3} \leq \frac{x}{\sin x} \leq \frac{\pi}{2}.$$

Step 2: Apply the comparison theorem for integrals

Theorem (Integral comparison). If $m \leq f(x) \leq M$ for all $x \in [a, b]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

The length of the interval is

$$b-a = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi - \pi}{6} = \frac{\pi}{3}.$$

Applying the theorem with $m = \pi/3$, $M = \pi/2$, and $b-a = \pi/3$:

$$\frac{\pi}{3} \cdot \frac{\pi}{3} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{\pi}{2} \cdot \frac{\pi}{3}.$$

Step 3: Evaluate and compare the bounds

- Lower bound:

$$\frac{\pi}{3} \cdot \frac{\pi}{3} = \frac{\pi^2}{9}.$$

- Upper bound:

$$\frac{\pi}{2} \cdot \frac{\pi}{3} = \frac{\pi^2}{6}.$$

This gives us:

$$\frac{\pi^2}{9} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{\pi^2}{6}.$$

Since $\frac{\pi^2}{6} \leq \frac{2\pi^2}{9}$ (equivalent to $\frac{1}{6} \leq \frac{2}{9}$, i.e. $3 \leq 4$, which is true), the upper bound implies:

$$\int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{\pi^2}{6} \leq \frac{2\pi^2}{9}.$$

Conclusion

Combining the lower and upper bounds:

$$\boxed{\frac{\pi^2}{9} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{2\pi^2}{9}.}$$

□